

BFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend

Complete

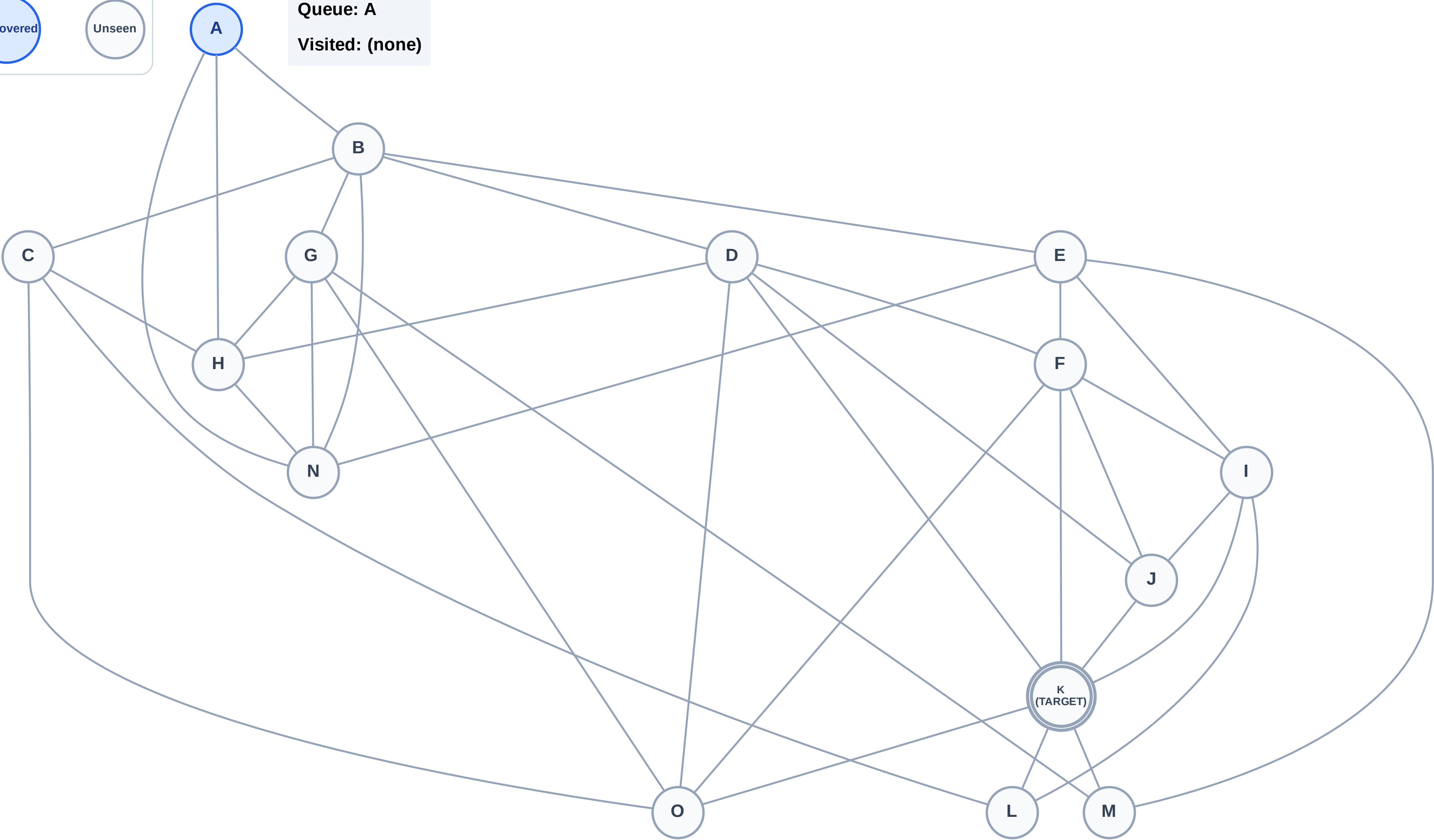
Processing

Discovered

Unseen

Queue: A

Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

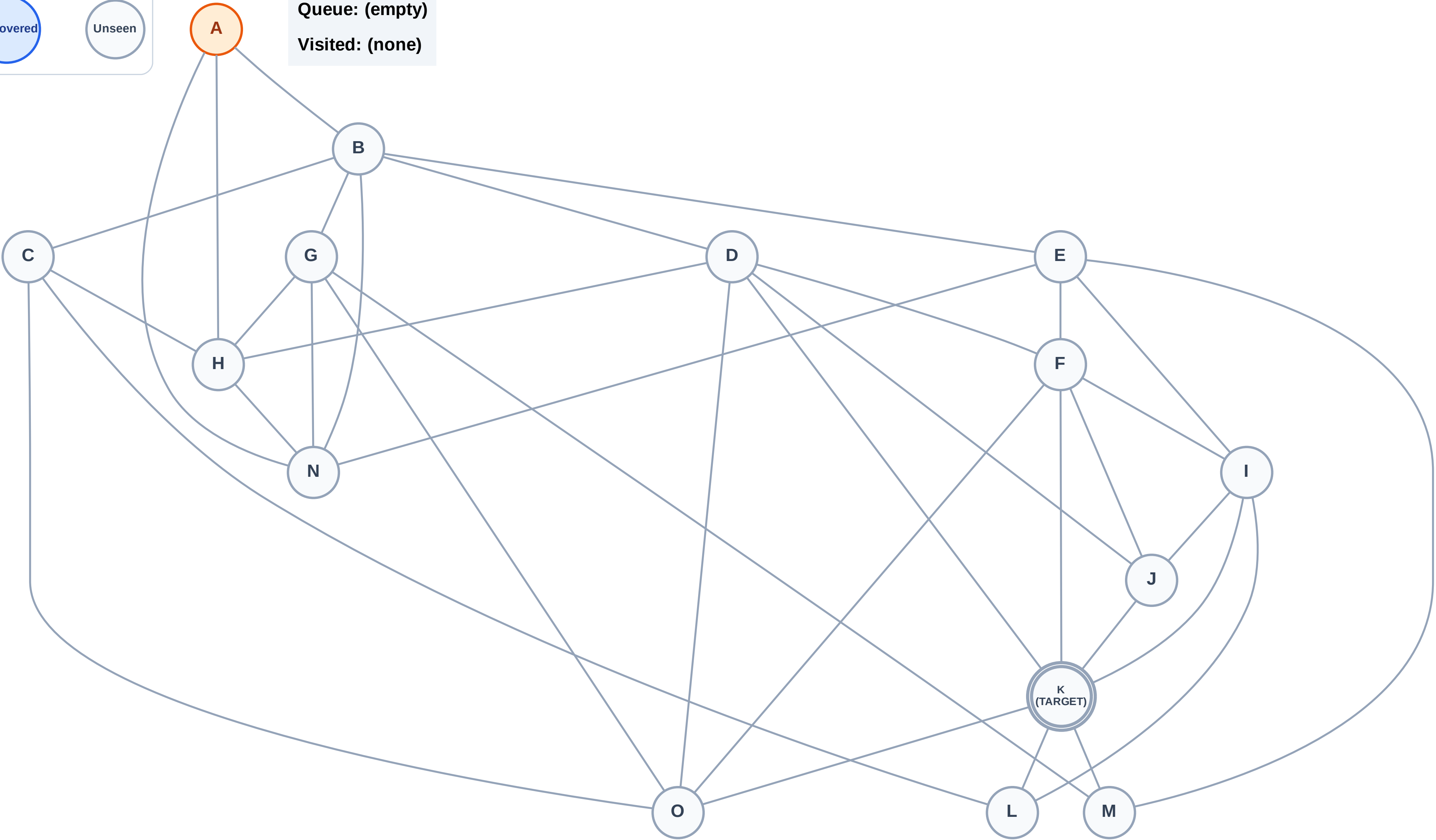
Complete

Processing

Discovered

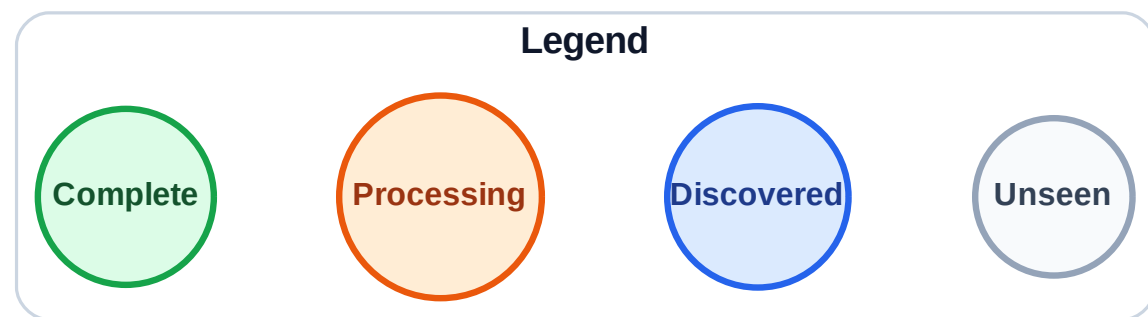
Unseen

Queue: (empty)
Visited: (none)

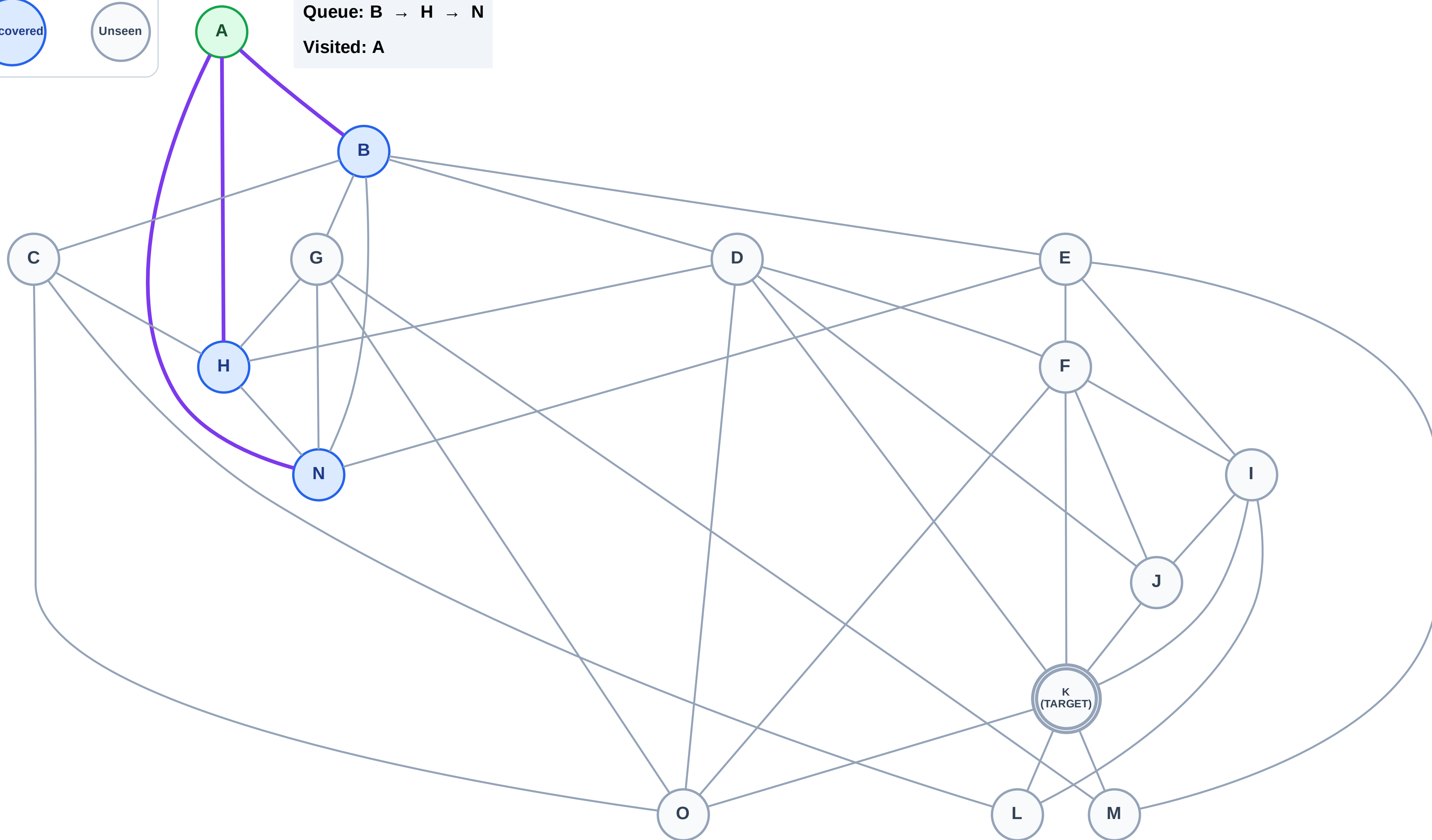


BFS Traversal

Step 3: Discover B, H, N from A



Queue: B → H → N
Visited: A



BFS Traversal

Step 4: Process node B

Legend

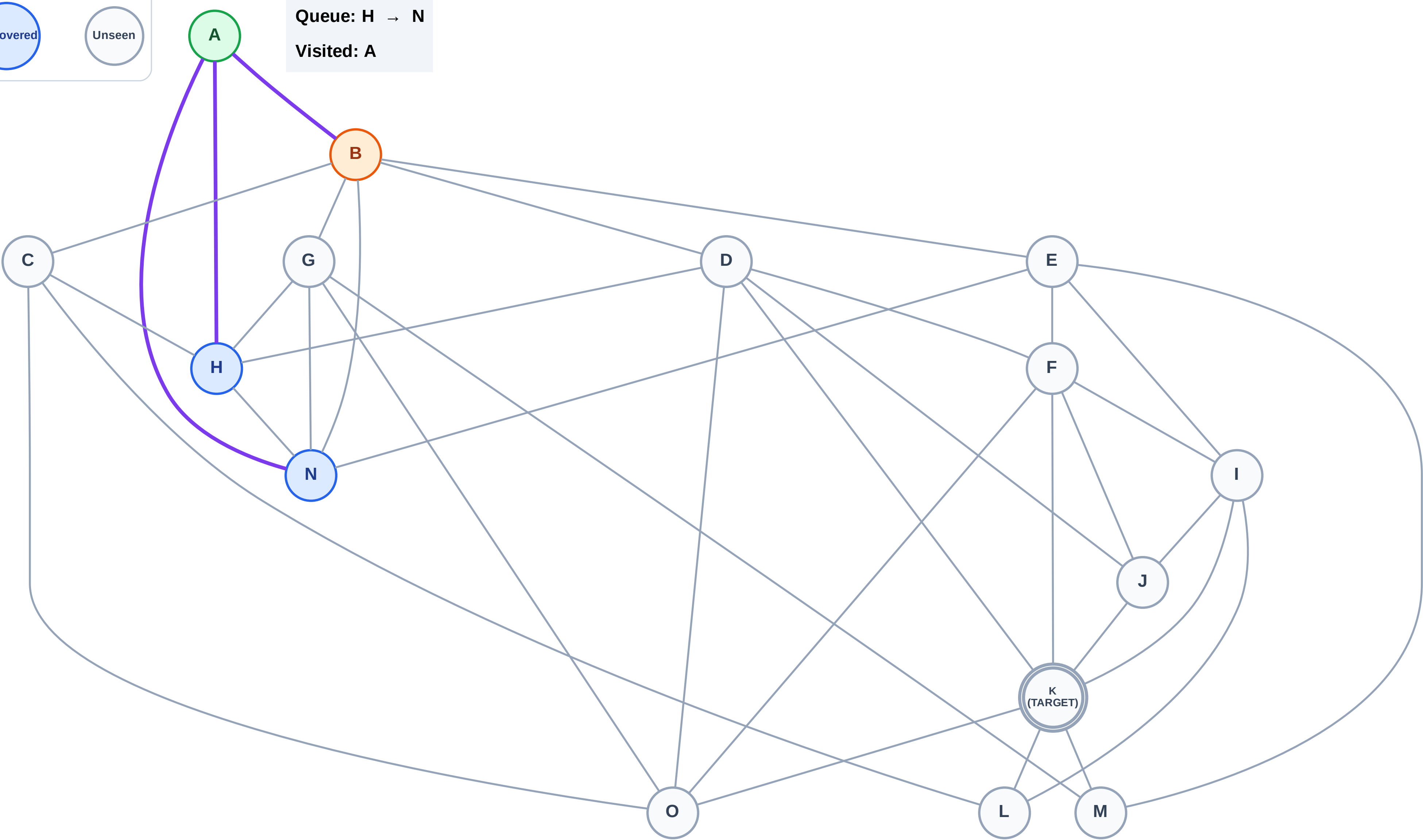
Complete

Processing

Discovered

Unseen

Queue: H → N
Visited: A



BFS Traversal

Step 5: Discover C, D, E, G from B

Legend

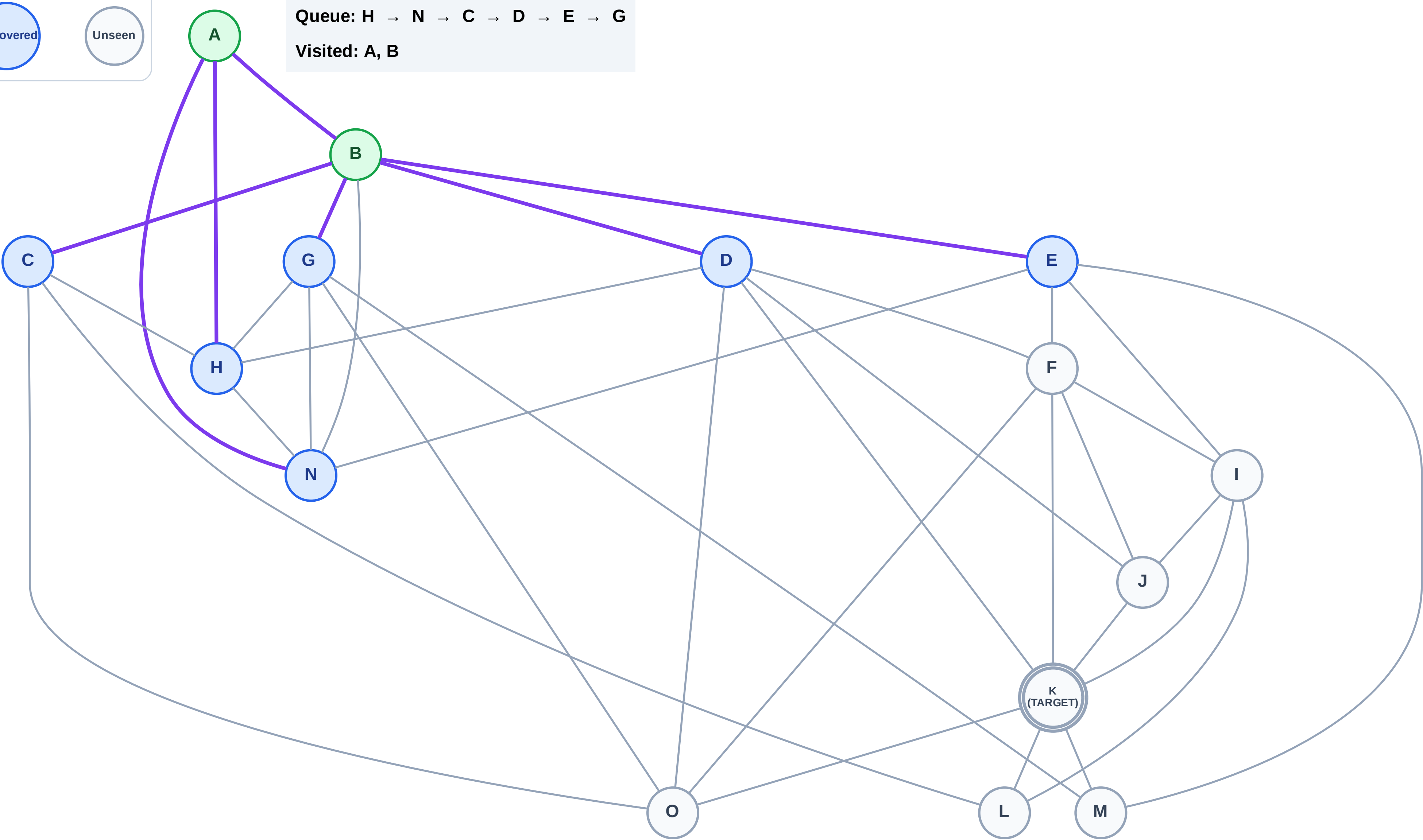
Complete

Processing

Discovered

Unseen

Queue: H → N → C → D → E → G
Visited: A, B



BFS Traversal

Step 6: Process node H

Legend

Complete

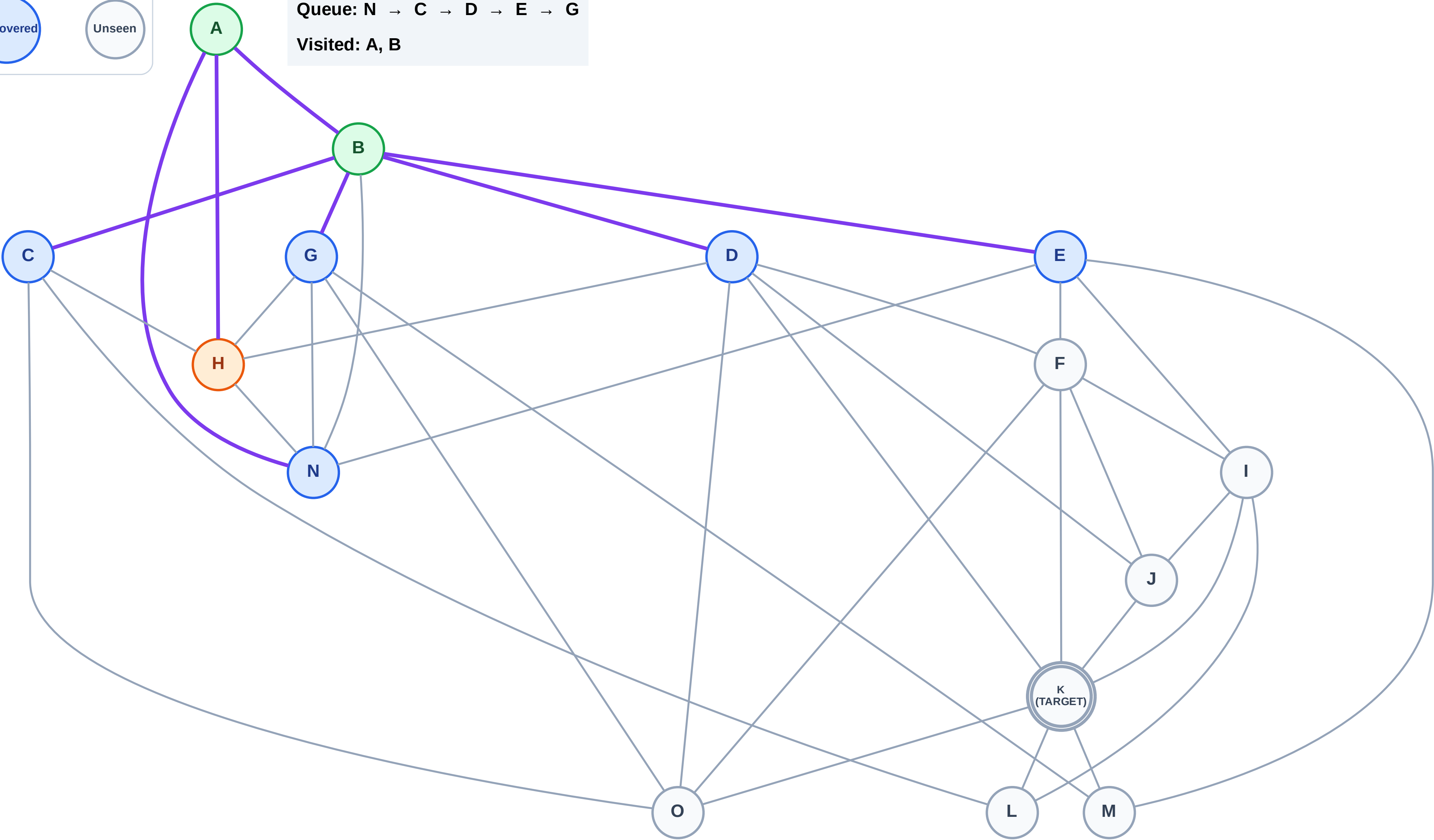
Processing

Discovered

Unseen

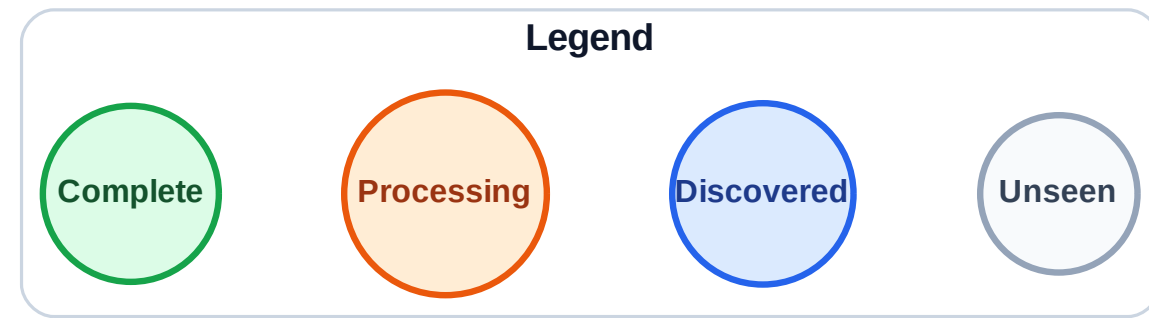
Queue: N → C → D → E → G

Visited: A, B

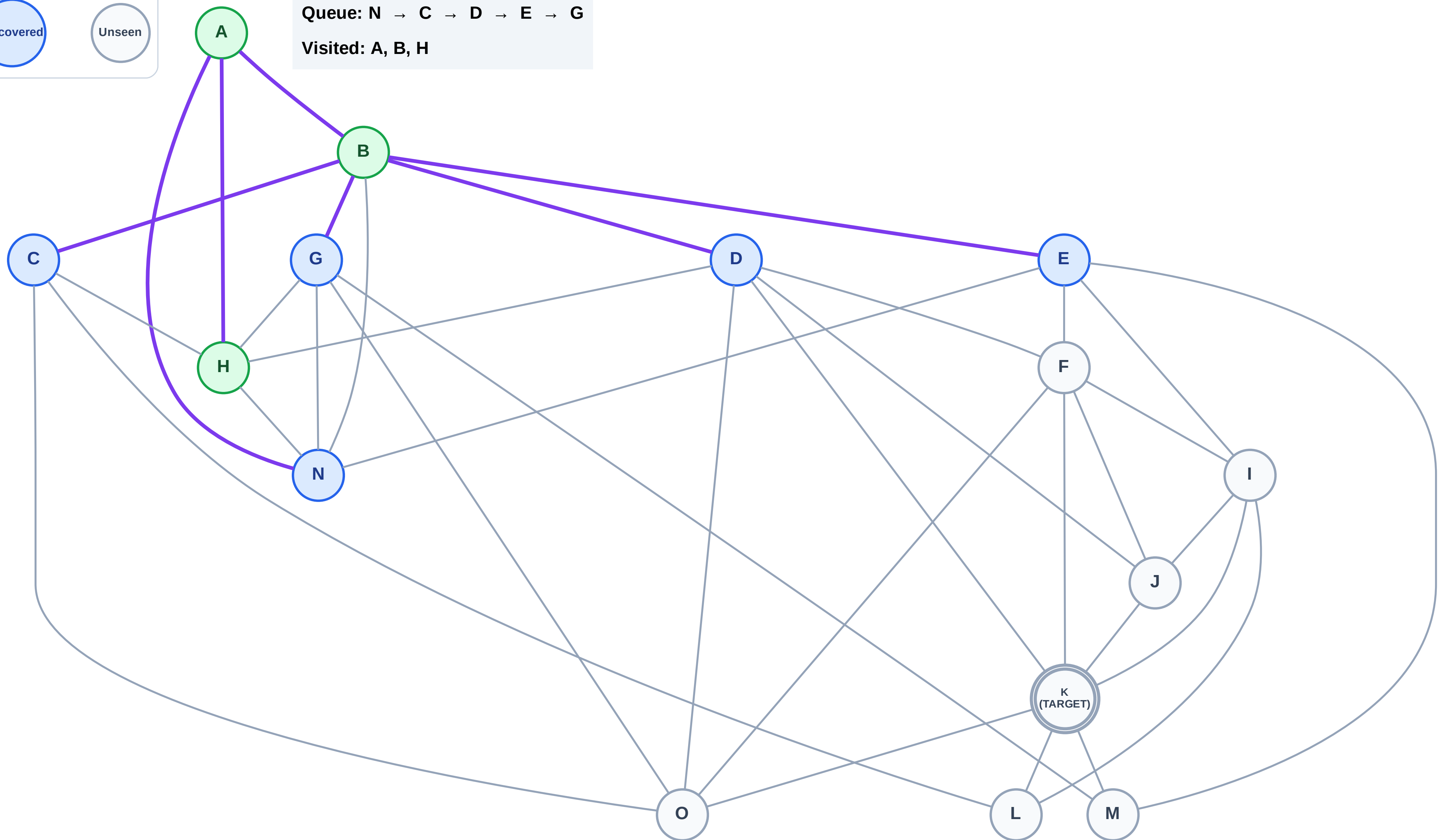


BFS Traversal

Step 7: No new neighbors from H



Queue: N → C → D → E → G
Visited: A, B, H



BFS Traversal

Step 8: Process node N

Legend

Complete

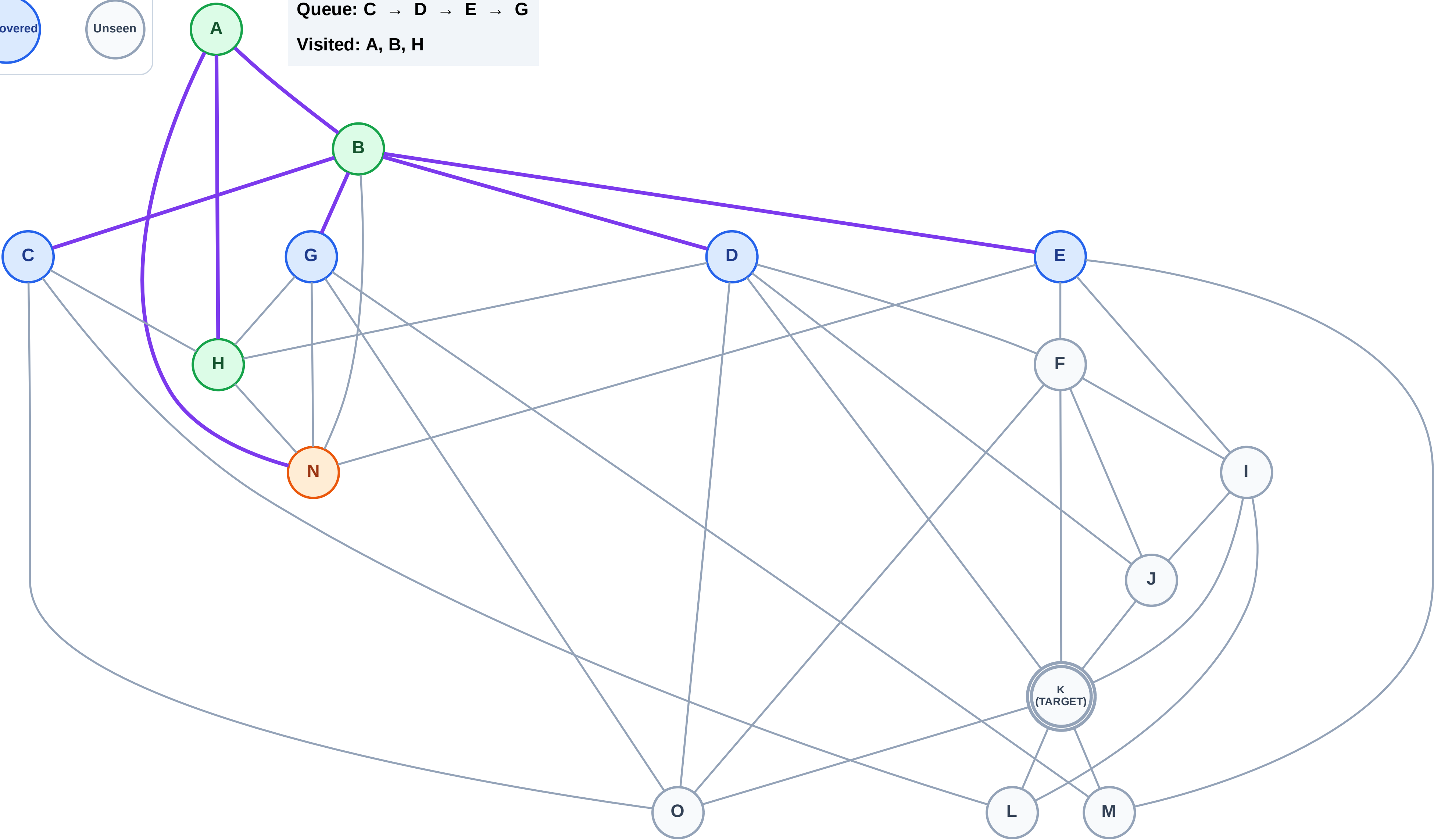
Processing

Discovered

Unseen

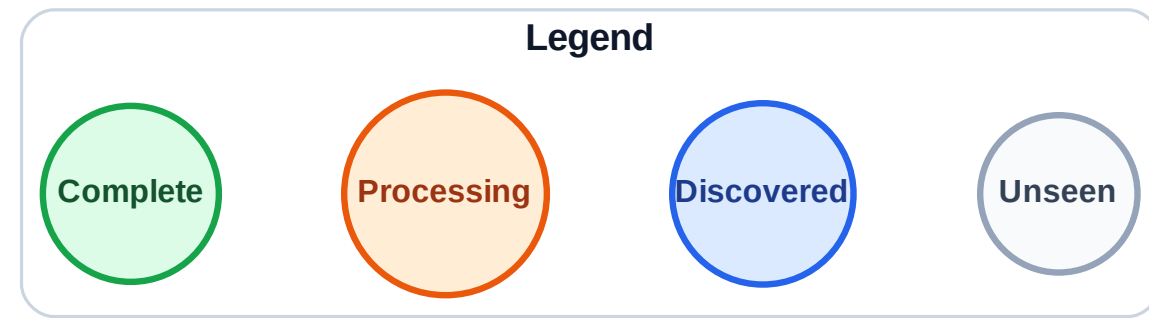
Queue: C → D → E → G

Visited: A, B, H



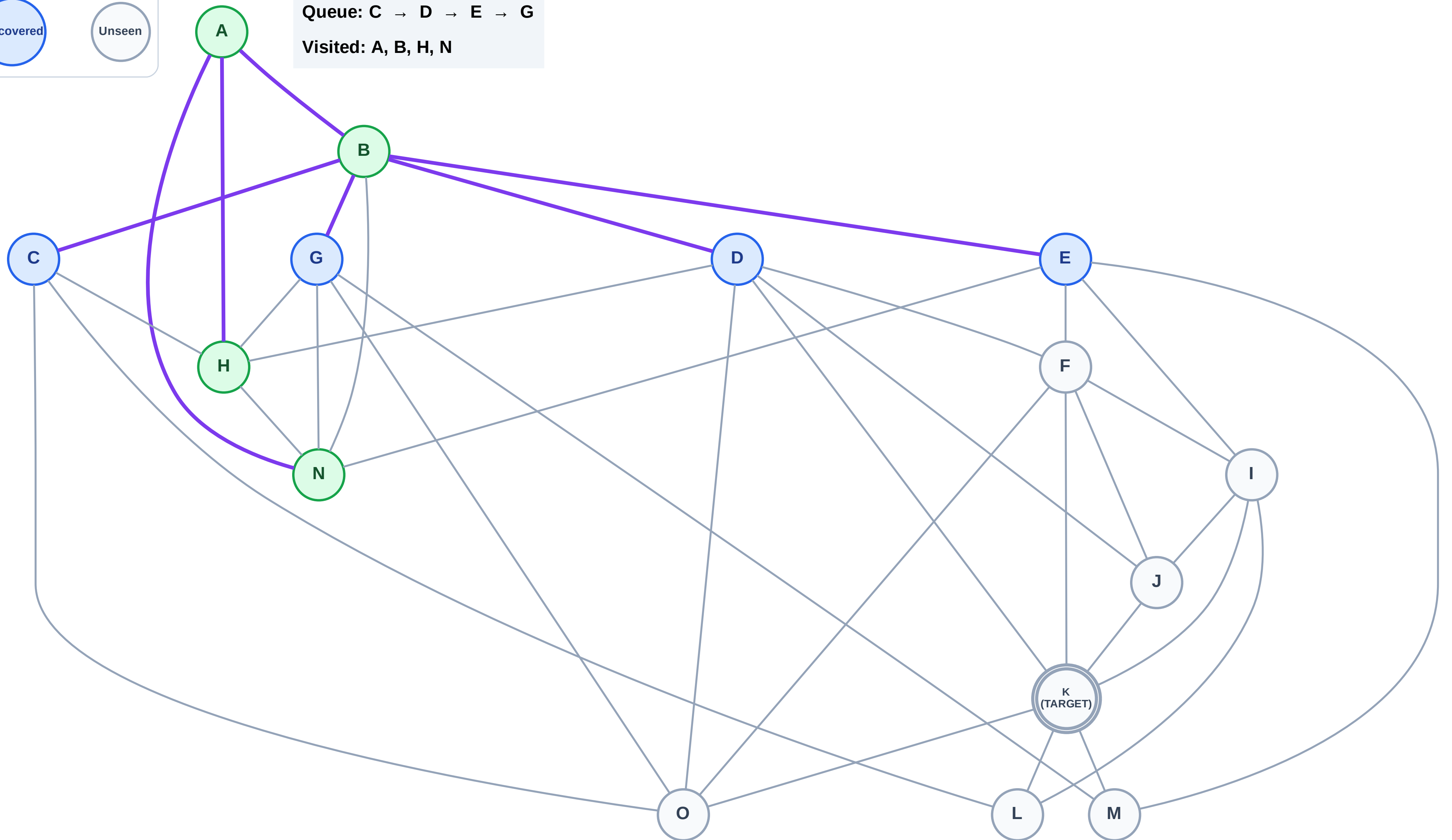
BFS Traversal

Step 9: No new neighbors from N



Queue: C → D → E → G

Visited: A, B, H, N



BFS Traversal

Step 10: Process node C

Legend

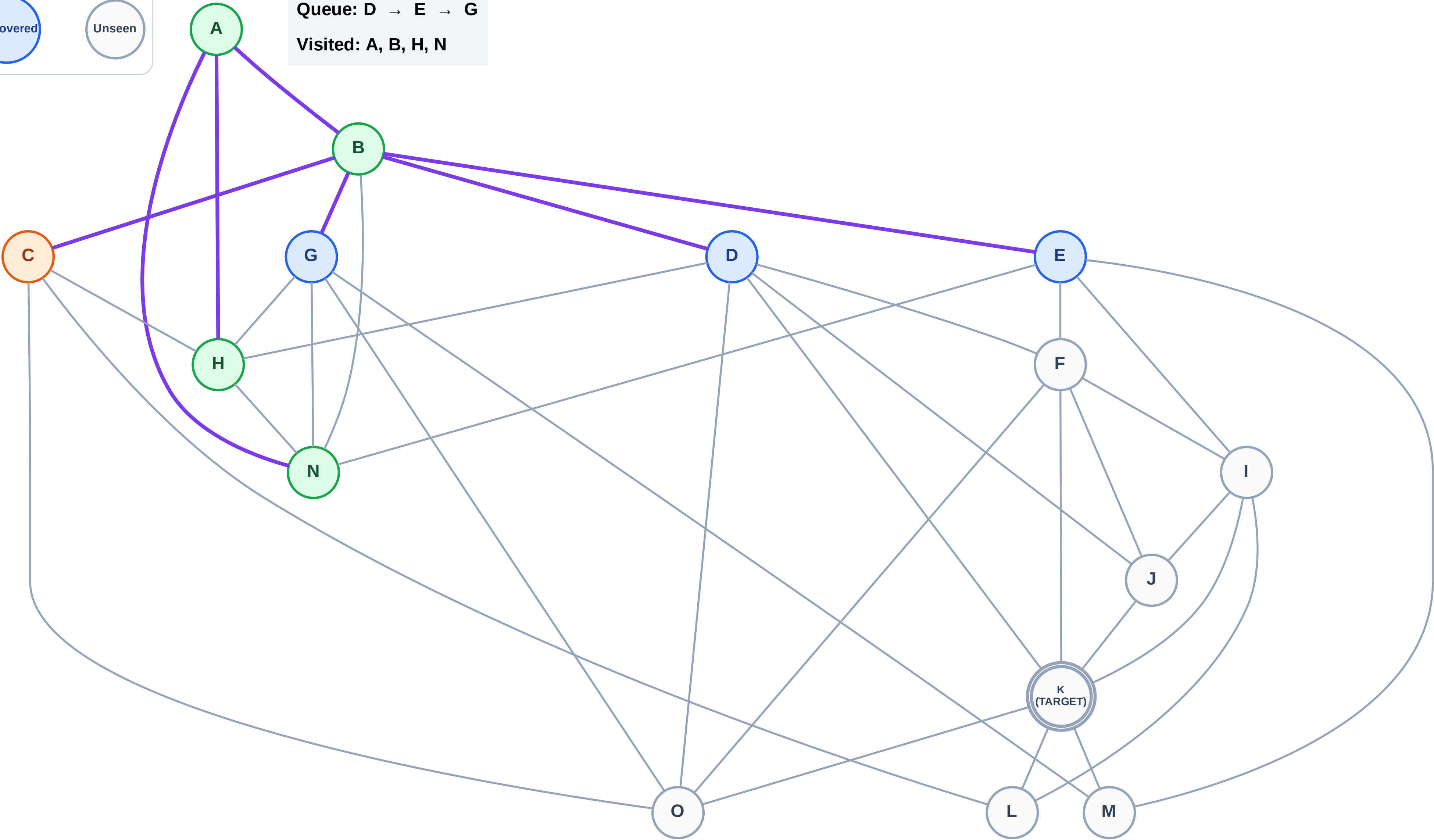
Complete

Processing

Discovered

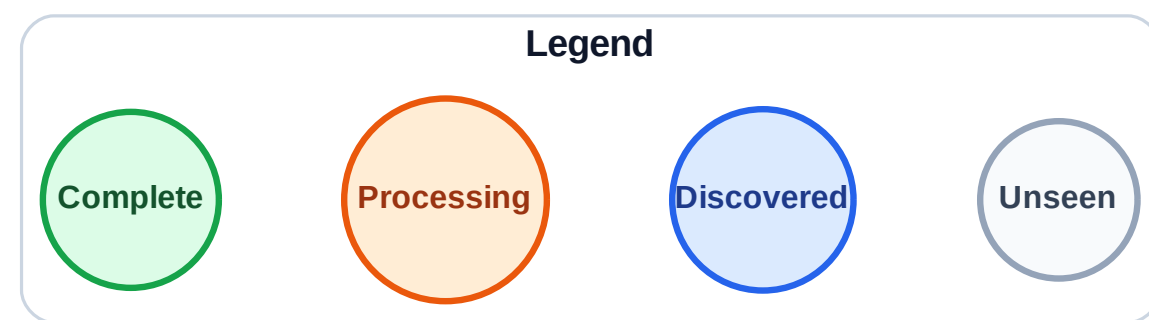
Unseen

Queue: D → E → G
Visited: A, B, H, N

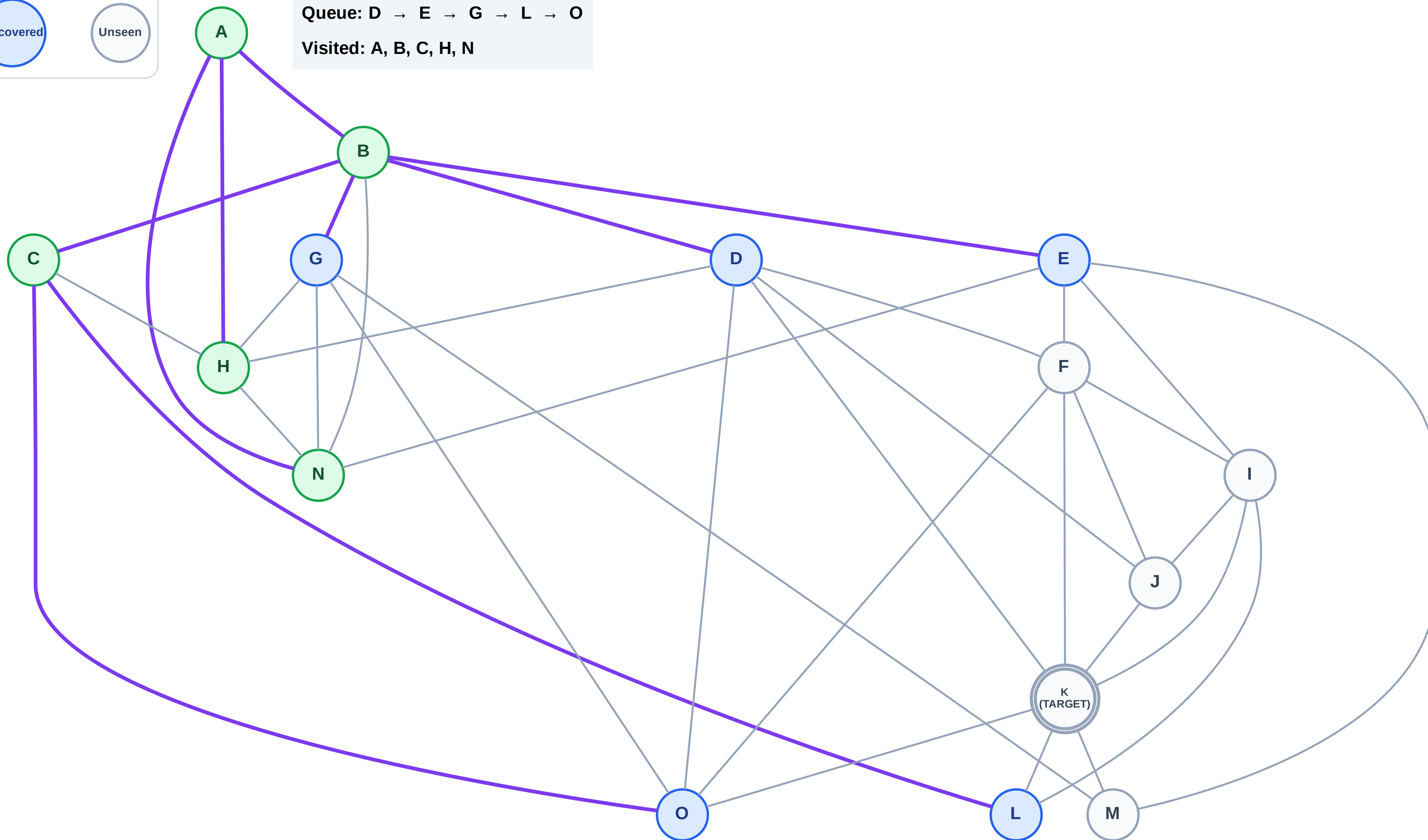


BFS Traversal

Step 11: Discover L, O from C



Queue: D → E → G → L → O
Visited: A, B, C, H, N



BFS Traversal

Step 12: Process node D

Legend

Complete

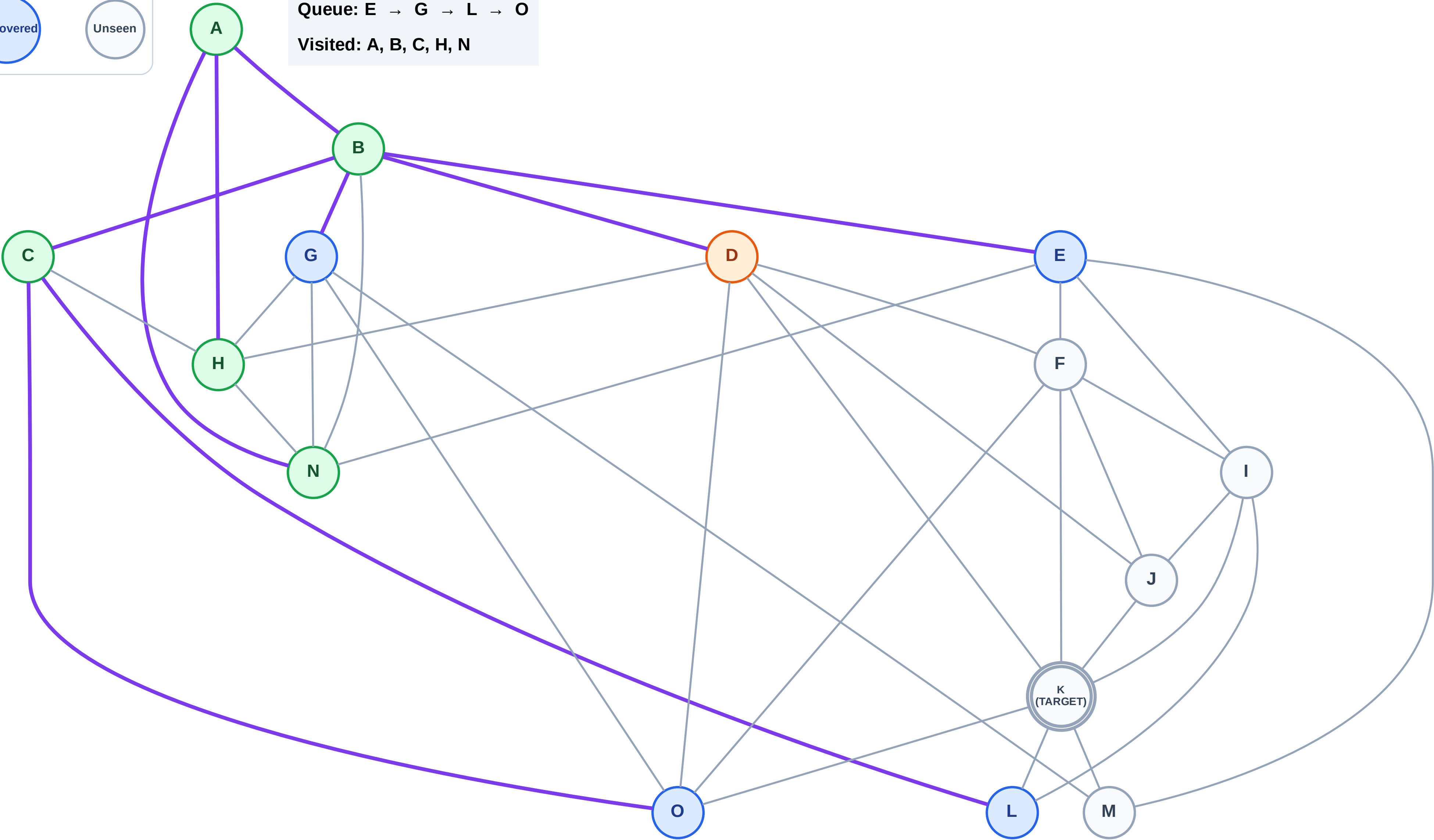
Processing

Discovered

Unseen

Queue: E → G → L → O

Visited: A, B, C, H, N



BFS Traversal

Step 13: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (3 hops): A → B → D → K

Legend

Complete

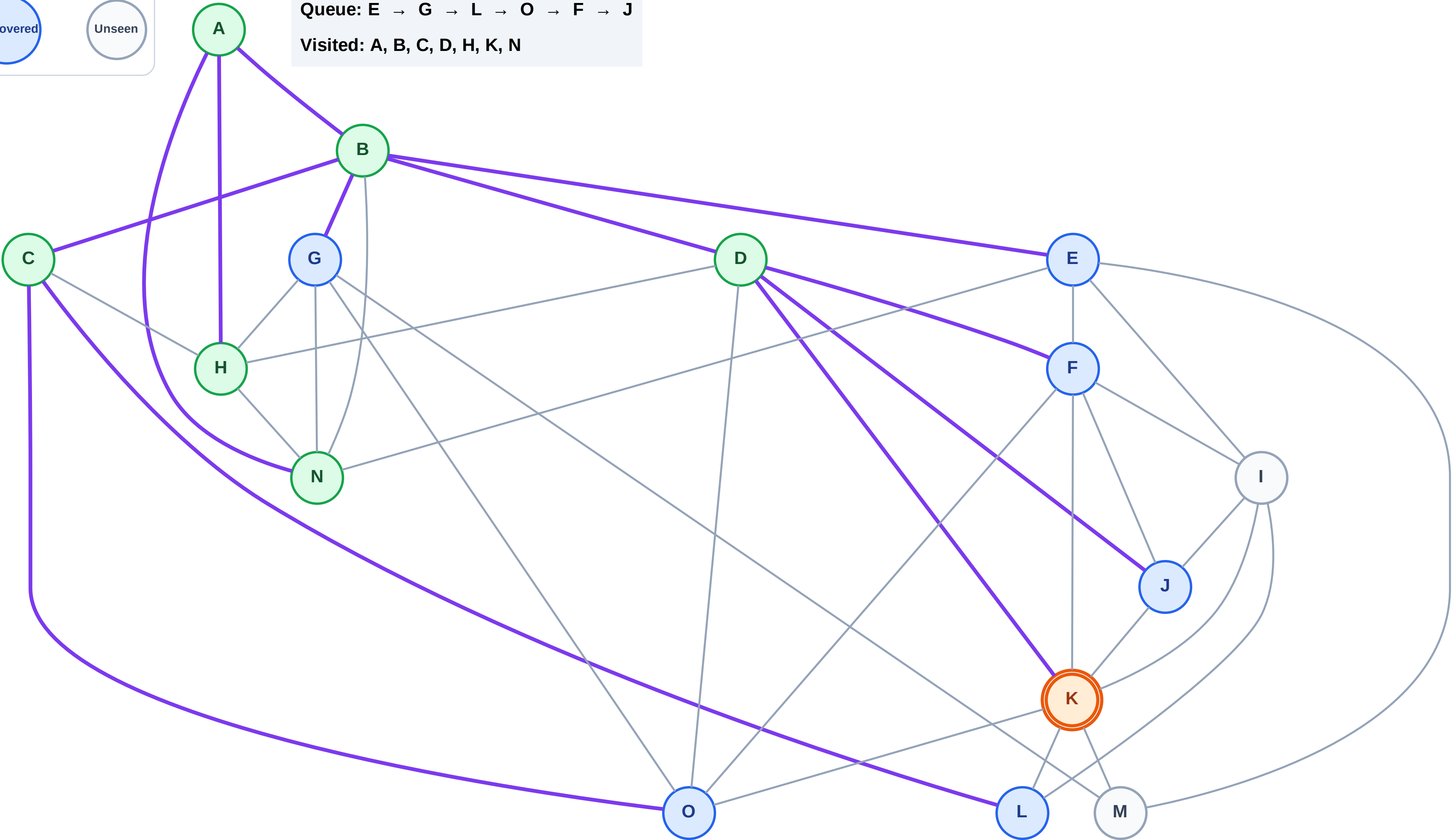
Processing

Discovered

Unseen

Queue: E → G → L → O → F → J

Visited: A, B, C, D, H, K, N



DFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend

Complete

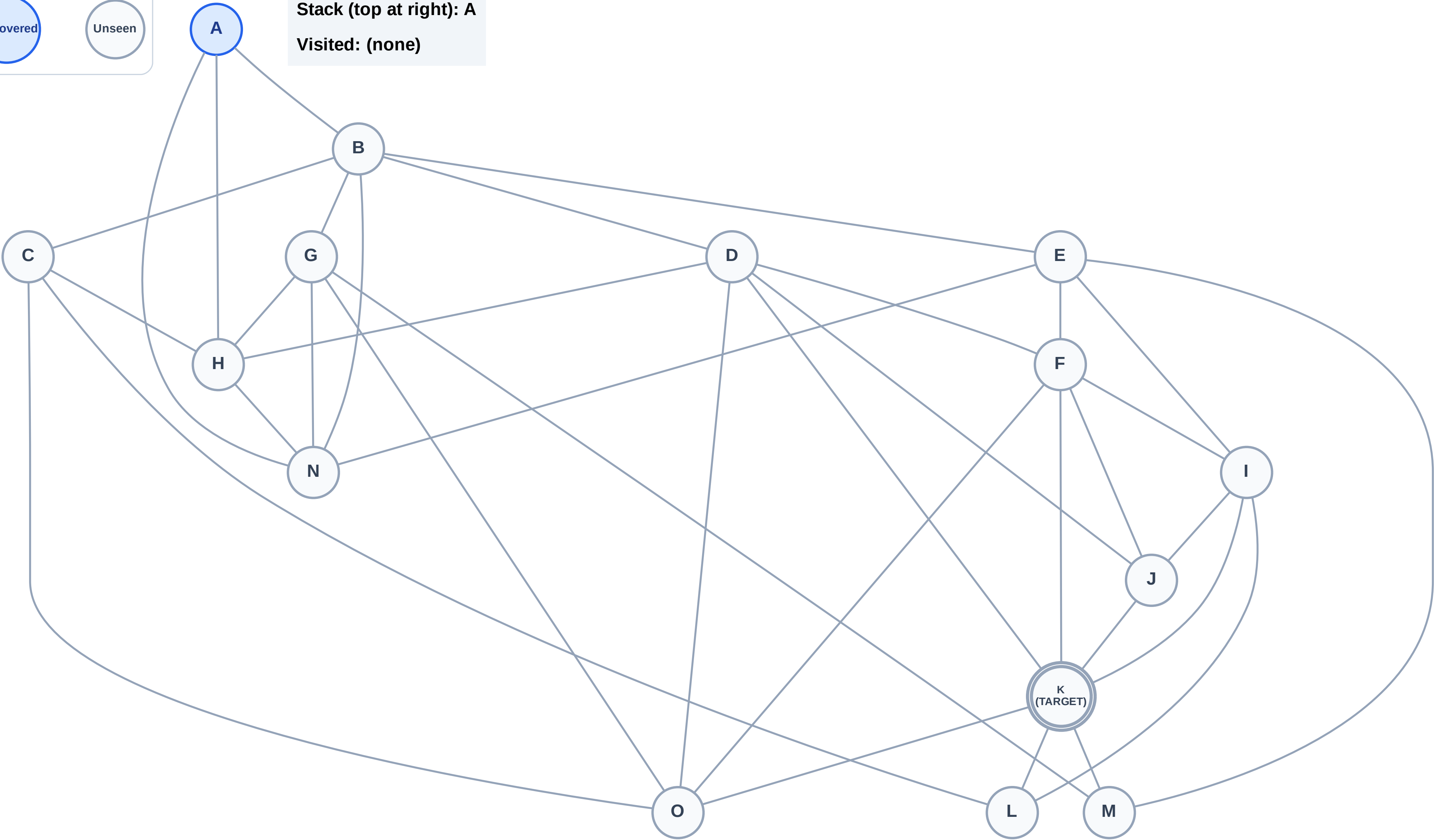
Processing

Discovered

Unseen

Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

Complete

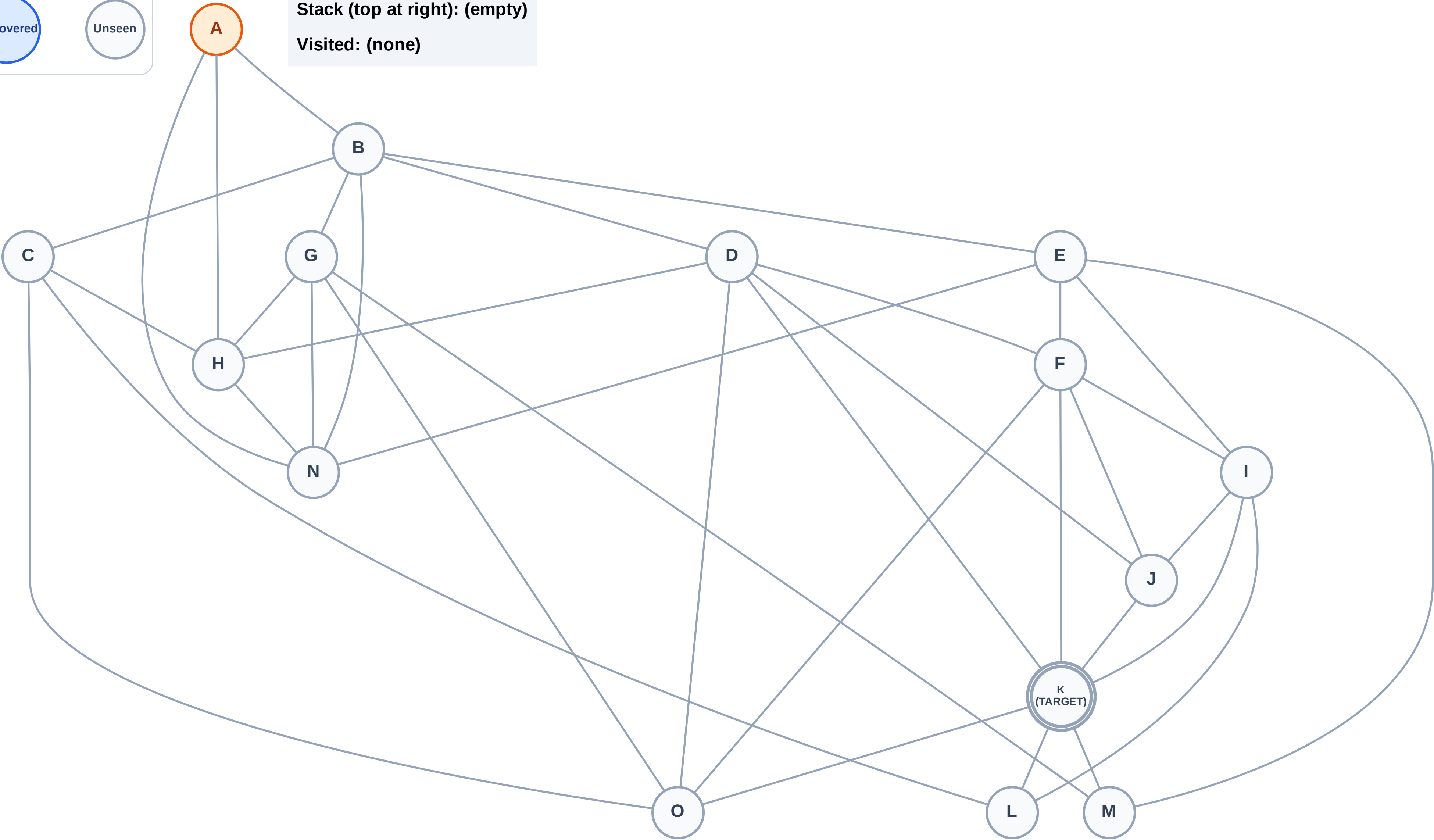
Processing

Discovered

Unseen

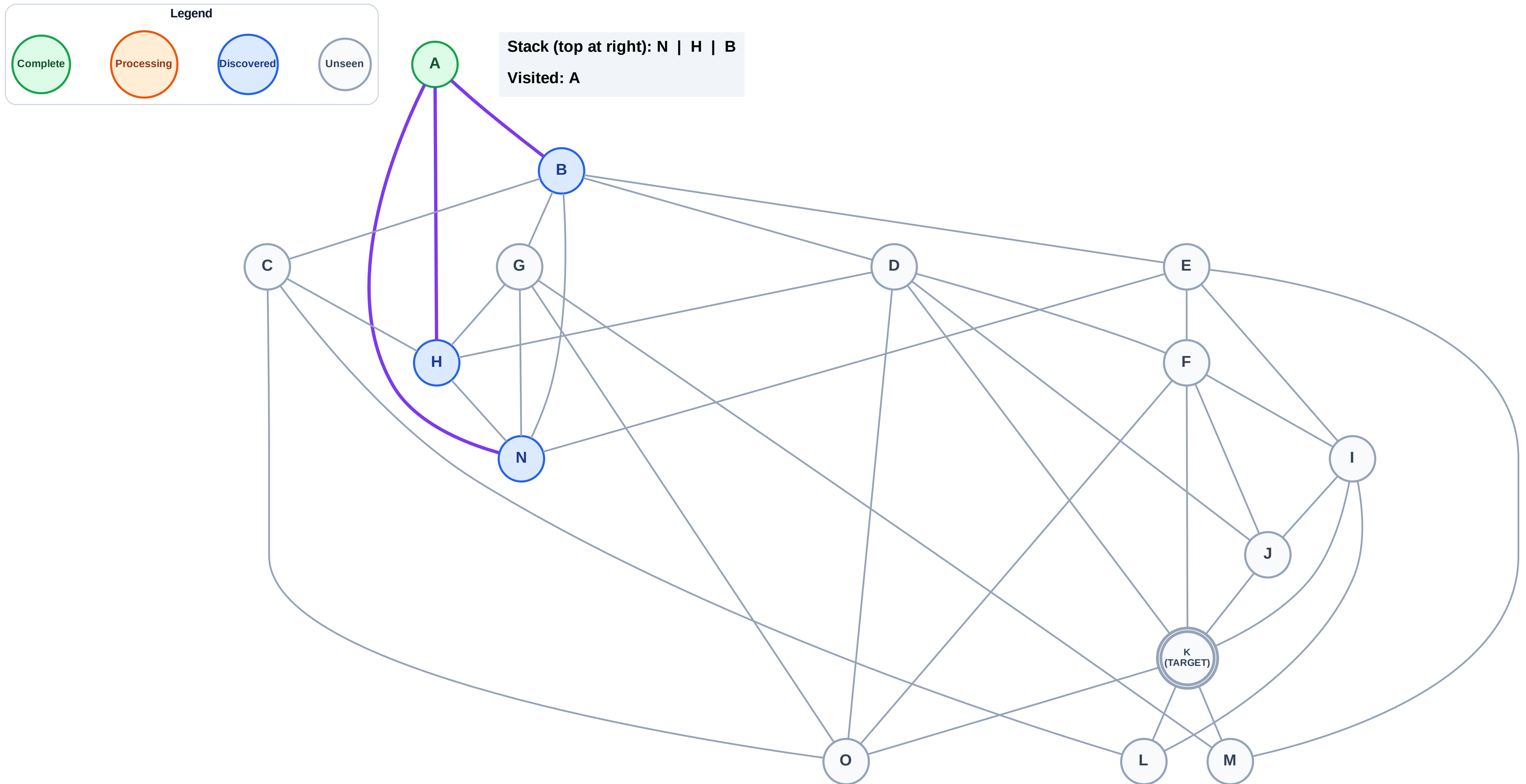
Stack (top at right): (empty)

Visited: (none)



Step 3: Discover N, H, B from A

Step 3: Discover N, H, B from A



DFS Traversal

Step 4: Process node B

Legend

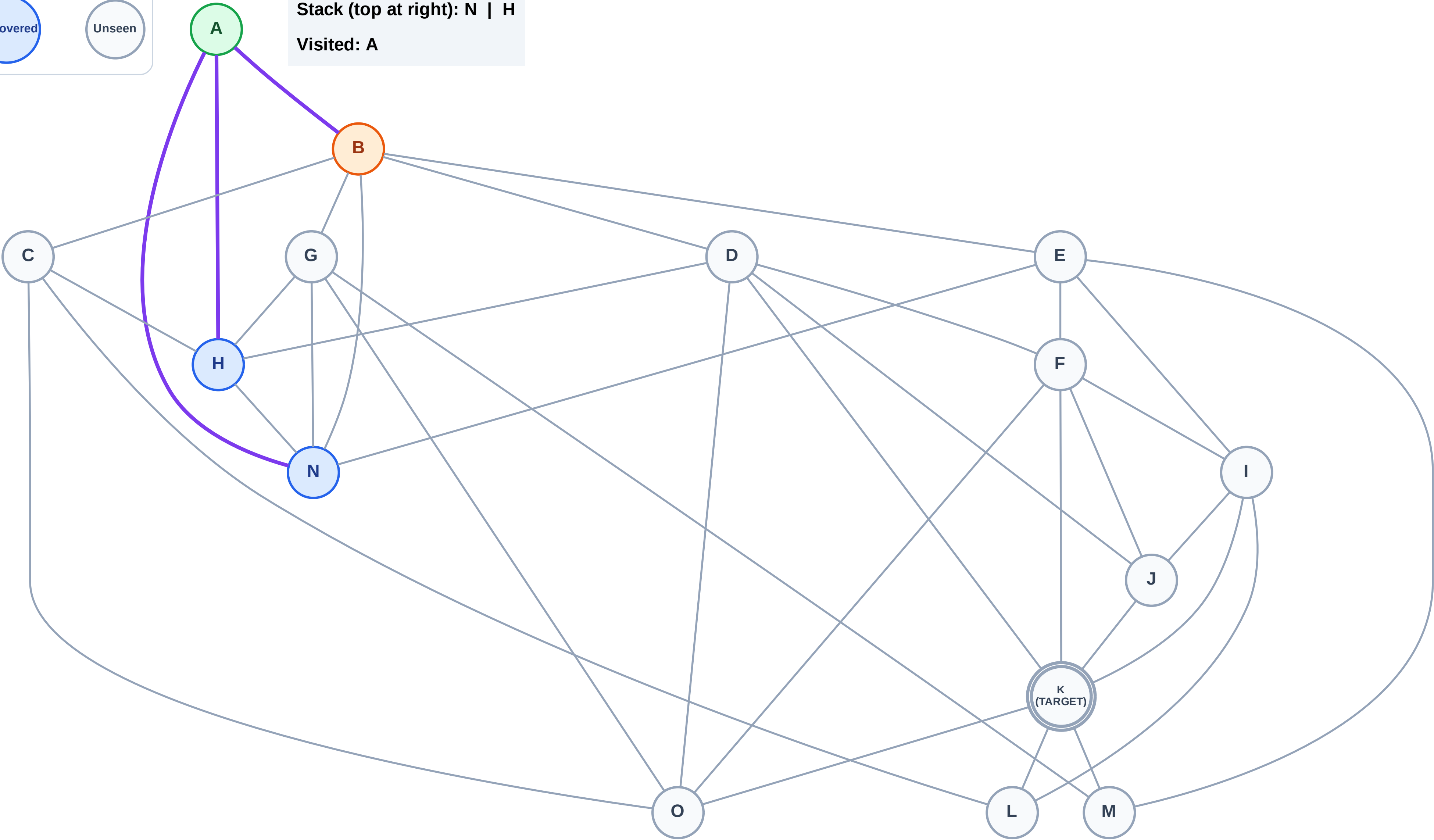
Complete

Processing

Discovered

Unseen

Stack (top at right): N | H
Visited: A



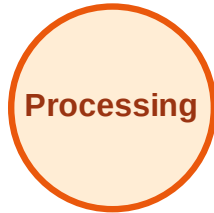
DFS Traversal

Step 5: Discover G, E, D, C from B

Legend



Complete



Processing



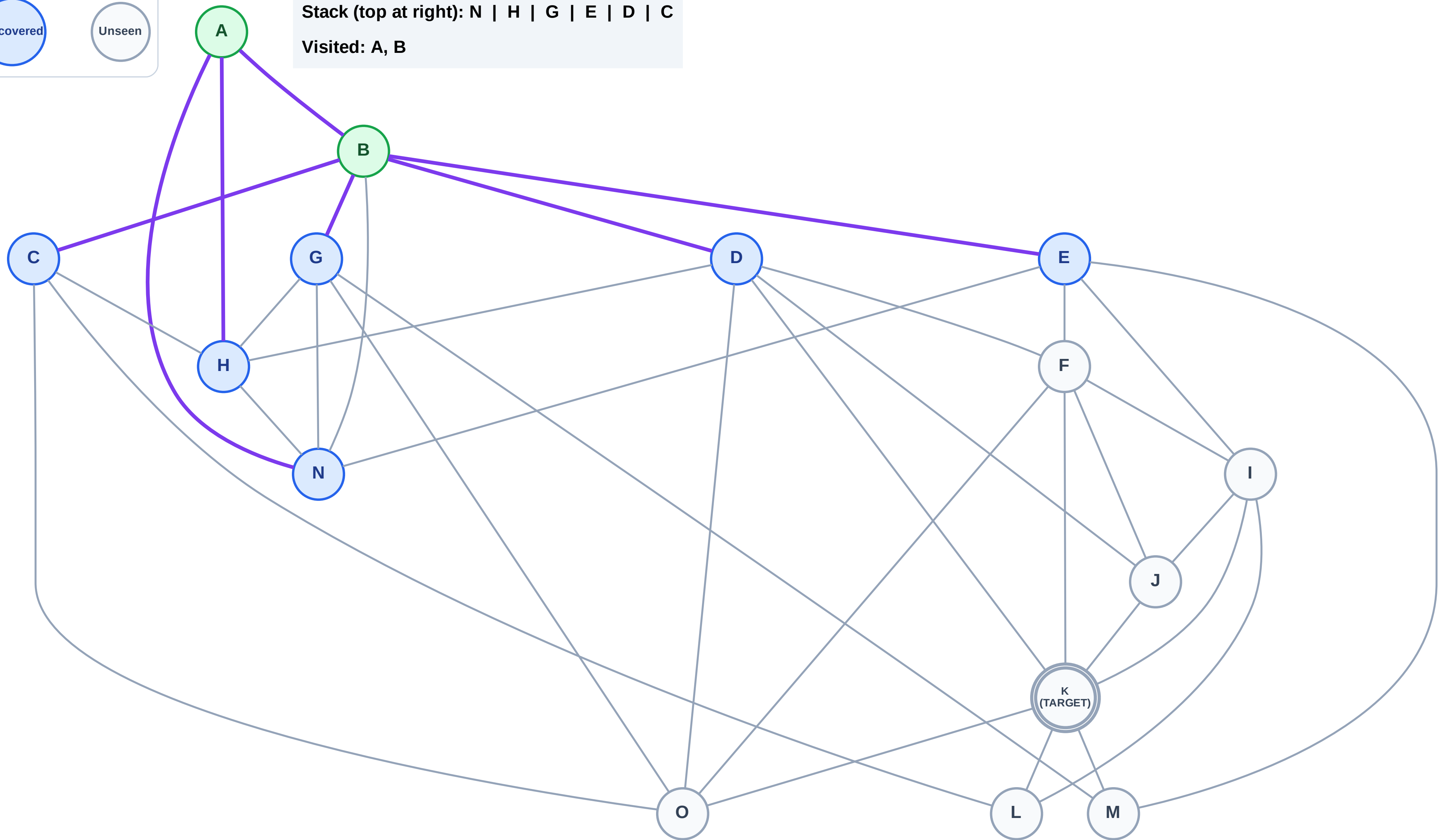
Discovered



Unseen

Stack (top at right): N | H | G | E | D | C

Visited: A, B



DFS Traversal

Step 6: Process node C

Legend

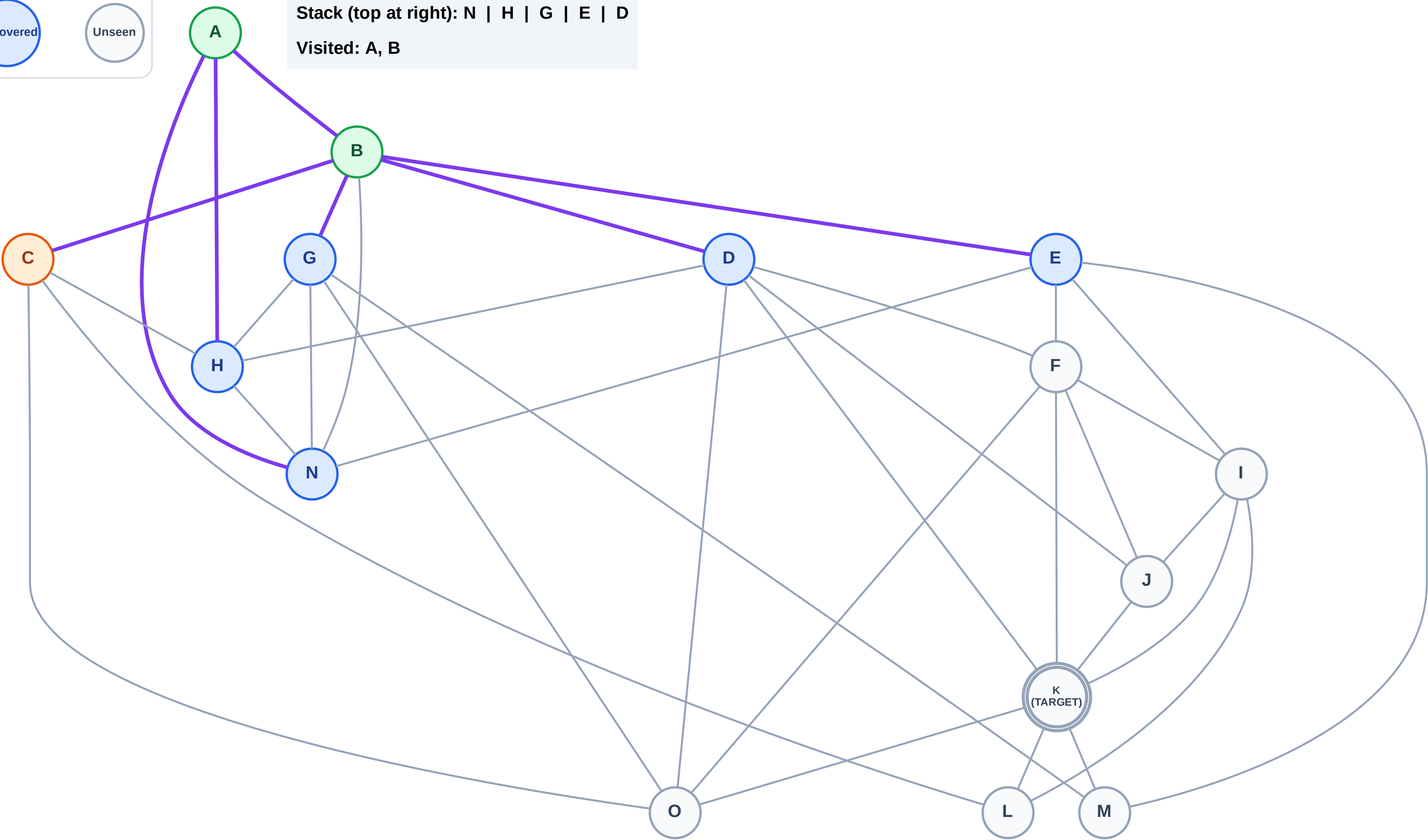
Complete

Processing

Discovered

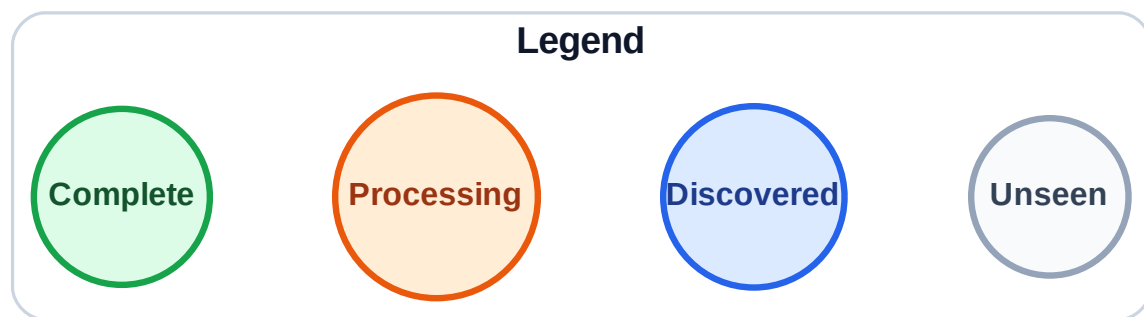
Unseen

Stack (top at right): N | H | G | E | D
Visited: A, B

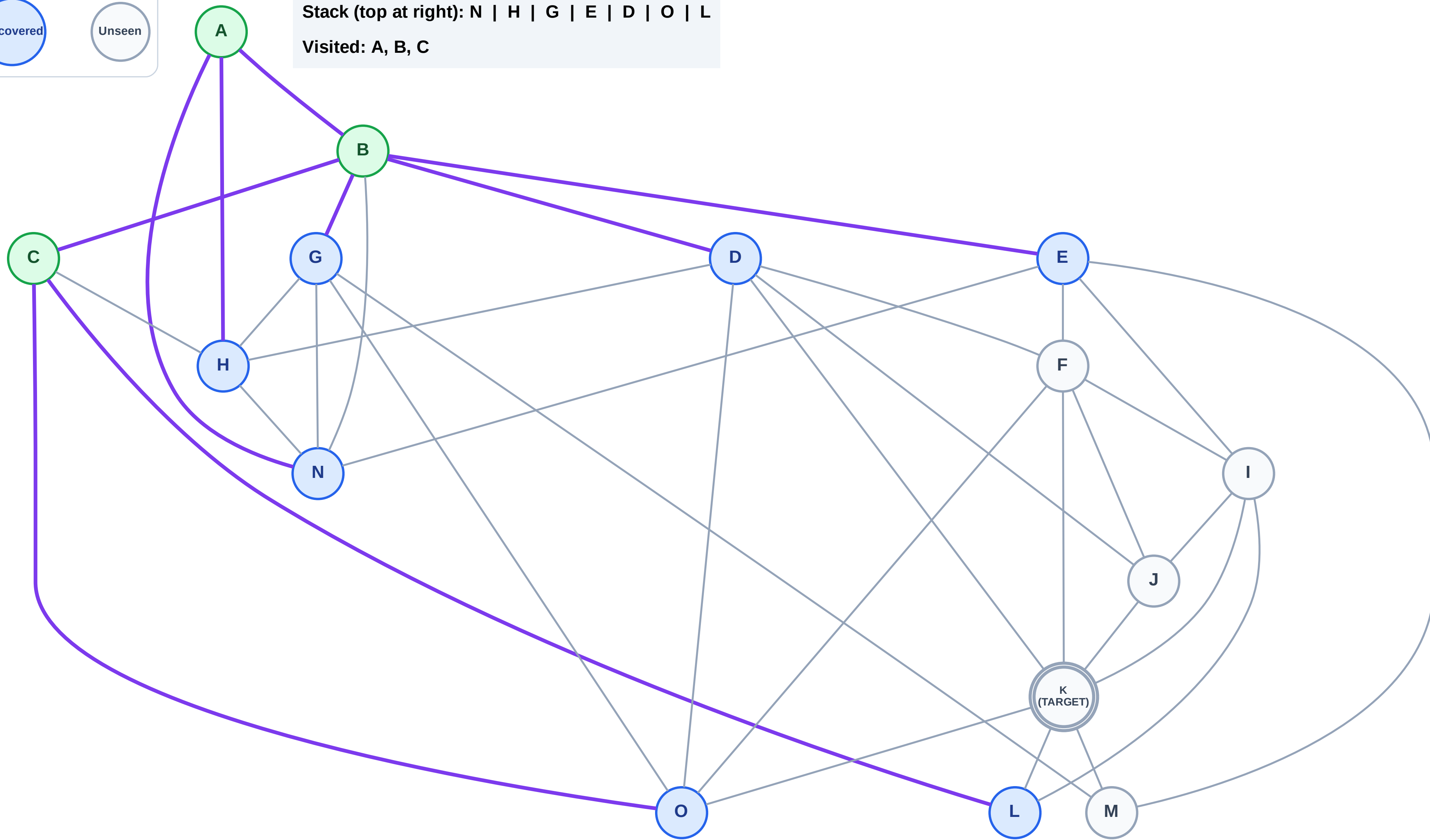


DFS Traversal

Step 7: Discover O, L from C



Stack (top at right): N | H | G | E | D | O | L
Visited: A, B, C



DFS Traversal

Step 8: Process node L

Legend

Complete

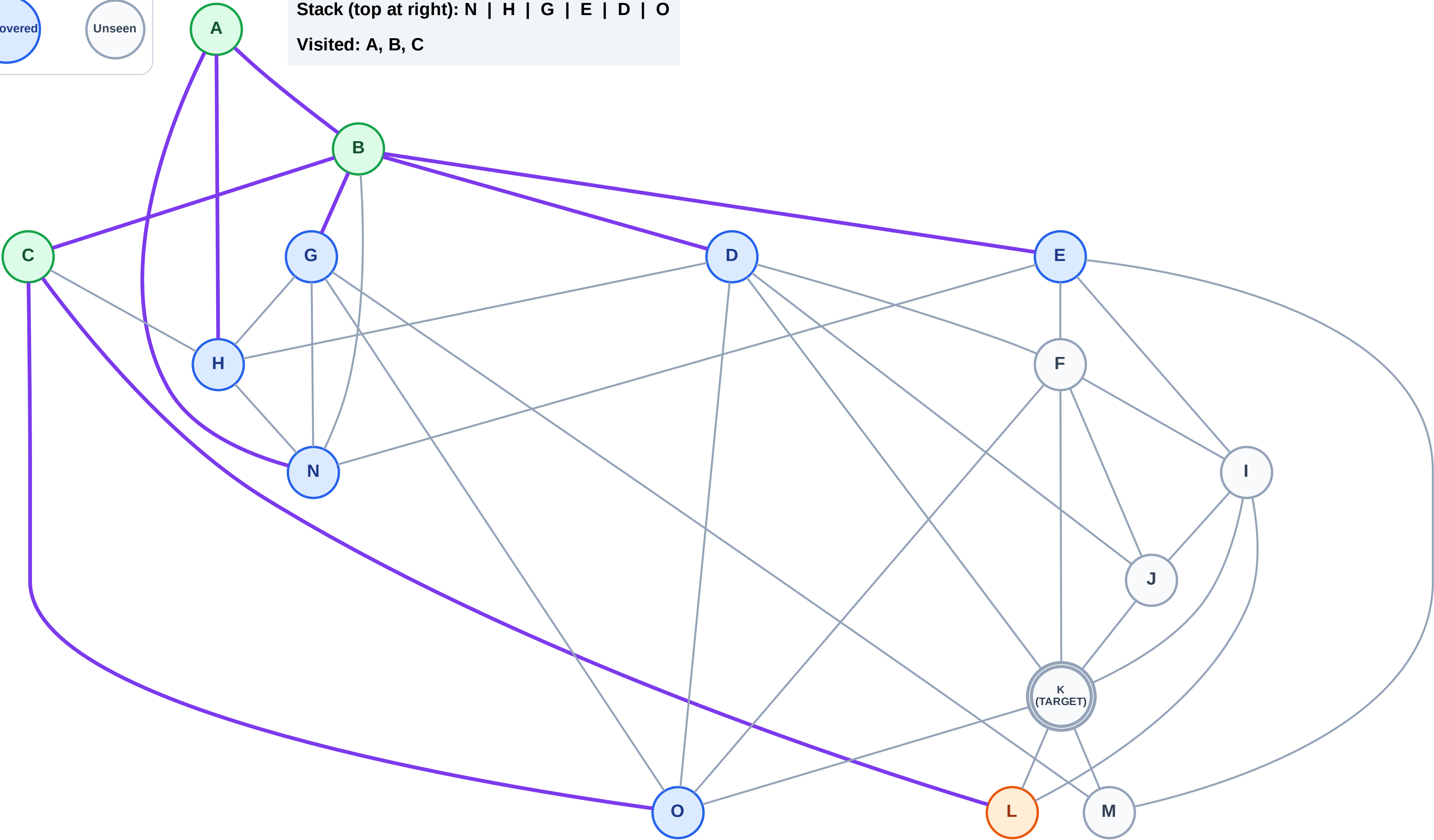
Processing

Discovered

Unseen

Stack (top at right): N | H | G | E | D | O

Visited: A, B, C



DFS Traversal

Step 9: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (4 hops): A → B → C → L → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): N | H | G | E | D | O

Visited: A, B, C, K, L

